Spec2Prop-InorgBench: An Open Raman–XRD Benchmark and AI-Assisted Screening Workflow for Inorganic Materials

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Abstract

Raman and X-ray diffraction (XRD) spectroscopies provide complementary vibrational and crystallographic fingerprints critical for inorganic material characterization. Despite the success of machine learning in materials informatics, the field lacks comprehensive, open benchmark datasets mapping raw spectral files to chemical families and material properties. We introduce Spec2Prop-InorgBench, a curated open-data Raman–XRD benchmark comprising 4,118 usable Raman spectra, 4,027 clean inorganic samples, and 1,844 property-matched samples, including 1,462 Raman–XRD linked pairs. To bridge the gap between raw instrumental output and machine learning, we develop a runtime preprocessing pipeline that transforms raw spectra into a standardized 2048-point vector, followed by chemistry-aware feature extraction yielding 32 global-shape and 126 expert spectral descriptors. We propose an AI-assisted screening workflow, Spec2Prop-Edge, which employs a LightGBM-DART classifier to predict nine major inorganic chemical families (e.g., silicates, oxides, carbonates) with an accuracy of 67.22% and a macro-F1 score of 56.57%. We extend this approach to spectra-to-property prediction, achieving macro-F1 scores of 61.09% for band-gap class, 67.56% for metal/non-metal classification, and 78.91% for formation-energy class. Spec2Prop-Edge is designed as a confidence-aware screening assistant for candidate prioritization and preliminary material-family interpretation, rather than replacing final experimental confirmation.

Keywords: Inorganic materials, Raman spectroscopy, X-ray diffraction, Materials informatics, Spectral preprocessing, LightGBM, Spectra-to-property prediction

1 Introduction

Raman spectroscopy serves as a powerful non-destructive analytical technique that provides a unique vibrational fingerprint of materials, making it indispensable in modern chemistry and materials science [1–5]. When combined with X-ray diffraction (XRD), which offers precise crystallographic and structural fingerprinting [6–12], researchers obtain a comprehensive multimodal view of inorganic materials. Such techniques are particularly critical for the identification and characterization of fundamental inorganic chemical families, including silicates, oxides, carbonates, sulfates, phosphates, sulfides, halides, and borates [13–15].

However, translating raw spectral files into actionable chemical insights presents significant challenges. Raw Raman and XRD spectra often suffer from nonuniform wavenumber or 2θ axes, baseline drift due to fluorescence, instrumental noise, arbitrary intensity scaling, duplicate measurements, and invalid rows [16, 17]. While machine learning (ML) has seen remarkable success in materials informatics [18–22], the majority of spectral ML research either relies on proprietary datasets, focuses exclusively on pre-cleaned idealized spectra, or treats the model as a "black box" without chemical interpretation [23–28]. There is a pressing need for an open, reproducible Raman/XRD benchmark that bridges the gap from raw spectral files to confidence-aware material screening.

To address this gap, we present **Spec2Prop-InorgBench**, a curated open-data inorganic Raman–XRD benchmark. Our contribution is not merely a dataset, but a complete workflow encompassing raw file parsing, runtime spectral preprocessing, and chemistry-aware feature extraction. We demonstrate the utility of this benchmark through **Spec2Prop-Edge**, an AI-assisted inorganic material family screening tool powered by a LightGBM-DART classifier [29, 30]. Furthermore, we extend our workflow to spectra-to-property prediction, including band-gap, metallicity, and formation-energy classifications [31–37].

The remainder of this manuscript is organized as follows: Section 2 details the dataset construction, preprocessing, feature extraction, and ML modeling. Section 3 presents the family classification and property prediction results, along with chemical interpretations of spectral features. Section 4 discusses the limitations of the current approach, and Section 5 concludes the paper.

2 Materials and Methods

2.1 Open spectral and materials-property resources

The foundation of Spec2Prop-InorgBench relies on the aggregation and harmonization of diverse open-access data resources. The primary source for raw Raman and XRD spectra is the RRUFF database [2], which provides unedited instrumental files.

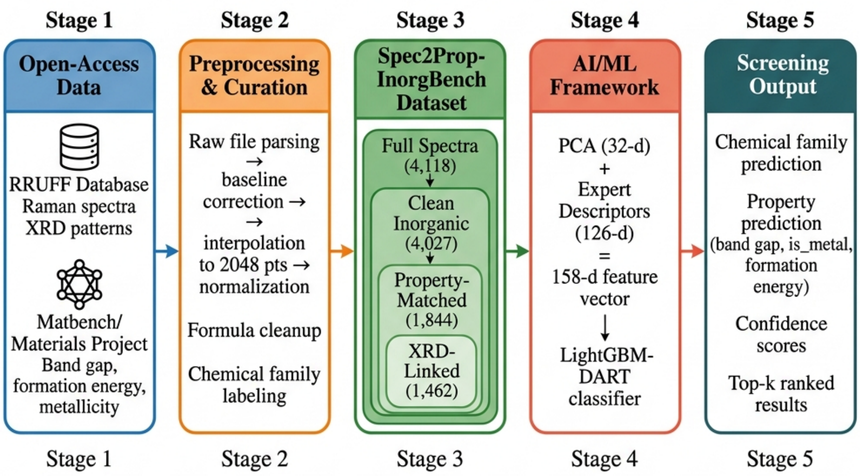

Fig. 1: Overall Spec2Prop-InorgBench workflow. Open Raman and XRD spectral resources are curated, formula-normalized, linked with inorganic family and materials-property labels, transformed into model-ready spectral representations, and used for confidence-aware screening.

Material properties, including band-gap and formation energy, were linked via the Materials Project [19] and Matbench [18]. Table 1 summarizes these sources.

2.2 Construction of Spec2Prop-InorgBench

We developed a rigorous data curation pipeline to construct the benchmark dataset. Raw records underwent formula normalization to resolve chemical ambiguities. As summarized in Table 2 and illustrated in Figure 2, filtering yielded 4,118 usable Raman spectra, 4,027 clean inorganic samples, and 1,844 property-matched samples. We successfully identified 1,462 samples possessing both Raman and XRD modalities, providing a rich multimodal subset for future research.

Table 1: Open data sources used for benchmark construction

Source	Data type	Main fields used	Linkage key	Role in benchmark	St
RRUFF [2]	Raman	Raw instrumental files (.txt)	RRUFF ID	Primary input feature	Int
RRUFF [2]	XRD	Powder diffraction patterns	RRUFF ID	Multimodal feature	Int
RRUFF [2]	Metadata	Chemistry, mineral name	RRUFF ID	Family ground truth	Int
Materials Project [19]	Properties	Band gap, formation energy	Chemical formula	Downstream targets	Int
Spec2Prop-InorgBench	Aggregated	Cleaned Raman/XRD + targets	Dataset index	Final release	Av

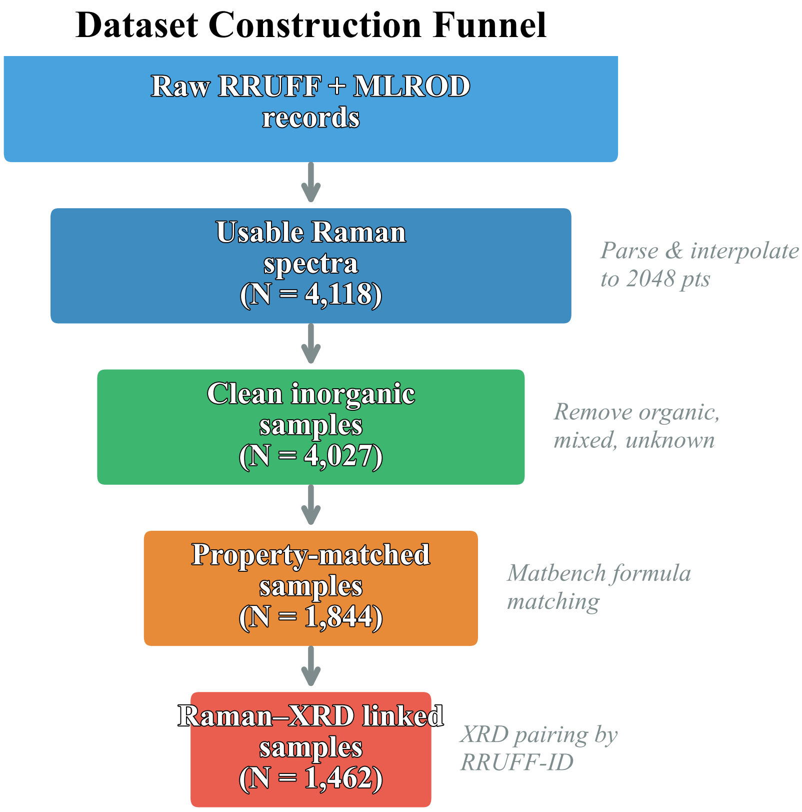

Fig. 2: Dataset construction funnel: Raw records → 4,118 usable Raman spectra → 4,027 clean inorganic samples → 1,844 property-matched samples → 1,462 Raman–XRD linked samples.

Table 2: Dataset subset statistics for Spec2Prop-InorgBench

Subset	Samples	Raman	XRD	Property labels	Main use
Full spectra	4,118	Yes	Partial	Partial	Pretraining, exploration
Clean inorganic	4,027	Yes	Partial	Partial	Family classification
Property-matched	1,844	Yes	Partial	Yes	Spectra-to-property
Raman–XRD linked	1,462	Yes	Yes	Partial	Multimodal fusion

2.3 Inorganic chemical-family label construction

Samples were categorized into nine major inorganic chemical families based on their anionic groups and dominant chemical structure (Table 3). These classes include Silicate, Oxide, Carbonate, Sulfate, Phosphate, Sulfide, Halide, Borate, and an Other/Rare category for minority groups.

Table 3: Inorganic chemical family definitions and spectral relevance

Family	Chemical meaning	Representative spectral features	Model role
Silicate	Contains $[SiO_4]^{4-}$ units	Strong Si–O stretching (800–1200 cm^{-1})	Target Class
Oxide	Contains O^{2-} with metals	M–O vibrations (broad, variable)	Target Class
Carbonate	Contains $[CO_3]^{2-}$ units	Sharp C–O symmetric stretch ($\sim$1090 cm^{-1})	Target Class
Sulfate	Contains $[SO_4]^{2-}$ units	Sharp S–O symmetric stretch ($\sim$980–1010 cm^{-1})	Target Class
Phosphate	Contains $[PO_4]^{3-}$ units	P–O symmetric stretch ($\sim$960 cm^{-1})	Target Class
Sulfide	Contains S^{2-} with metals	Low wavenumber M–S modes ($<$500 cm^{-1})	Target Class
Halide	Contains F^-, Cl^-, Br^-, I^-	Very low wavenumber lattice modes	Target Class
Borate	Contains $[BO_3]^{3-}/[BO_4]^{5-}$	B–O stretching ($\sim$1300–1500 cm^{-1})	Target Class
Other/Rare	Nitrates, tungstates, etc.	Varies	Target Class

2.4 Runtime Raman and XRD spectral preprocessing

To enable ML models to consume disparate raw instrumental files, we implemented an automated runtime spectral preprocessing pipeline. The sequence of operations (Table 4) ensures baseline drift correction [38], noise reduction via Savitzky–Golay smoothing [17], and standardizes the spectral range to a uniform 2048-point vector.

2.5 Chemistry-aware spectral feature extraction

Rather than relying solely on black-box representations, we designed a chemistry-aware feature extraction module [23]. The 2048-point Raman vector is mapped to a highly compact 158-dimensional feature vector, comprising 32 PCA global-shape features and 126 expert spectral descriptors. This includes spectral moments, peak statistics, and diagnostic band ratios (Table 5).

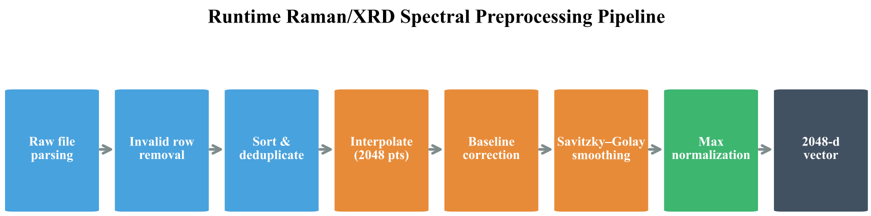

Fig. 3: Runtime Raman/XRD preprocessing pipeline: Raw file parsing → invalid row removal → sorting/deduplication → interpolation → baseline correction → smoothing → normalization → 2048-point vector.

Table 4: Runtime preprocessing operations for raw Raman/XRD files

Step	Operation	Chemistry/spectral reason	Output
1	Raw file parsing	Instruments export heterogeneous text formats	Array of (X, Y)
2	Invalid row removal	Cosmic rays and hardware errors yield NaNs	Cleaned array
3	Axis sorting	Wavenumber axes can be reversed	Ascending X
4	Duplicate removal	Multiple detector hits at same wavenumber	Unique X
5	Interpolation	Align to uniform grid (e.g., 200–1500 cm^{-1})	2048-d vector
6	Baseline correction	Remove fluorescence drift [38]	Flattened vector
7	Savitzky–Golay	Smooth high-frequency instrumental noise	Smoothed vector
8	Min–Max normalization	Correct intensity variance across instruments	$[0, 1]$ scaled vector

Table 5: Chemistry-aware spectral feature groups

Feature group	Dimension	Used in final model	Chemical interpretation
PCA global-shape features	32	Yes	Broad spectral background and continuum
Expert descriptors	126	Yes	Specialized extraction of diagnostic peaks
Peak statistics	TBD	Yes	Identification of primary vibrational modes
Spectral moments	TBD	Yes	Overall spectral mass and skewness
Diagnostic band ratios	TBD	Yes	Relative intensities of anion groups
Compact shape features	TBD	Yes	Low-resolution baseline approximation
Total vector	158	Yes	

2.6 Machine-learning models

The primary classifier evaluated is LightGBM augmented with Dropout meets Multiple Additive Regression Trees (DART) [29, 30], which prevents over-specialization and improves robustness on spectral datasets. We benchmarked this against Logistic Regression, SVM, Random Forest, XGBoost [39–46], and neural models (SimpleCNN1D, RamanFormer1D). Model configurations are detailed in Table 6.

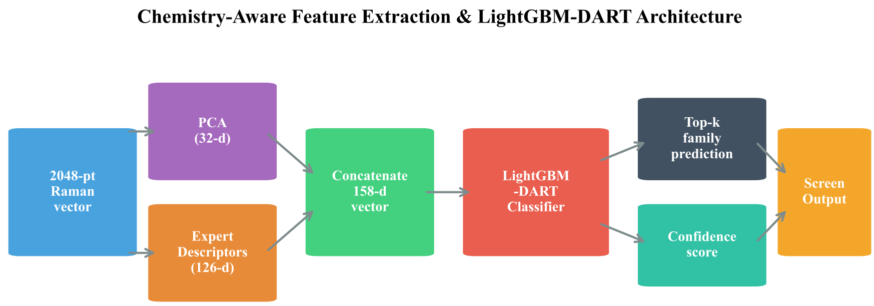

Fig. 4: Feature extraction and LightGBM-DART architecture: The 2048-point Raman vector is reduced to 32 PCA features and 126 expert descriptors, forming a 158-dimensional vector used by the LightGBM-DART model to predict the top-k chemical family and confidence score.

Table 6: Machine-learning model configurations

Model	Input representation	Output task	Purpose	Key settings
Logistic Regression	158-d feature vector	Family	Baseline	$C = 1.0$, L2 penalty
SVM	158-d feature vector	Family	Baseline	RBF kernel
Random Forest	158-d feature vector	Family	Baseline	500 trees
XGBoost	158-d feature vector	Family	Baseline	depth=6, lr=0.1
LightGBM-DART	158-d feature vector	Family/Properties	Main classifier	Dropout boosting
SimpleCNN1D	2048-d raw vector	Family	Deep learning	3 conv blocks
RamanFormer1D	2048-d raw vector	Family	Deep learning	Transformer-based
FusionSpec2PropNet	Multimodal features	Family	Advanced fusion	Concat layers
DualBranch Raman–XRD	2×2048-d vectors	Family	Multimodal baseline	Feature pooling

2.7 Leakage-safe evaluation protocol

To prevent data leakage caused by highly similar polymorphs or duplicated measurements of the same mineral deposit, we implemented a leakage-safe validation split strategy [47–50]. Table 7 outlines the evaluation protocol.

2.8 Confidence-aware deployment workflow

Our deployment architecture, Spec2Prop-Edge, incorporates confidence-calibration techniques [51] to report prediction uncertainty. Outputs are classified into High, Medium, and Low confidence tiers, functioning as an AI assistant for candidate prioritization rather than absolute identification.

Table 7: Leakage-safe evaluation protocol

Task	Dataset subset	Split strategy	Metrics	Leakage control
Family classification	Clean inorganic	80/20 Stratified	Acc, Macro-F1	Formula grouping
Property prediction	Property-matched	80/20 Stratified	Acc, Macro-F1	Formula grouping
Multimodal class.	Raman–XRD linked	80/20 Stratified	Acc, Macro-F1	Formula grouping
Confidence screening	Deployment set	Calibrated	Expected Cal. Error	Out-of-distribution tracking

3 Results and Discussion

3.1 Dataset composition and inorganic chemical-family distribution

The final Spec2Prop-InorgBench dataset reflects a natural imbalance typical in geological and synthetic materials databases. Silicates and oxides dominate the distribution, followed by carbonates and sulfates.

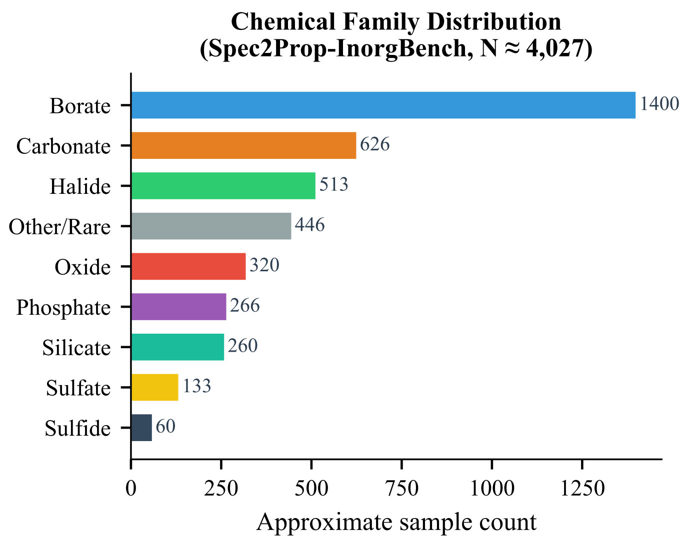

Fig. 5: Chemical family distribution of the Spec2Prop-InorgBench dataset.

3.2 Raman-based inorganic family screening

Table 8 presents the family classification performance across different model architectures. The LightGBM-DART model, leveraging the 158-d chemistry-aware feature vector, achieved an accuracy of 67.22% and a macro-F1 score of 56.57%. This performance demonstrates the efficacy of fusing expert descriptors with global PCA shapes for discriminating complex inorganic families.

Table 8: Inorganic chemical family classification results

Model	Input	Accuracy (%)	Macro-F1 (%)	Notes
Logistic Regression	158-d features	TBD	TBD	Linear baseline
SVM	158-d features	TBD	TBD	Non-linear baseline
Random Forest	158-d features	TBD	TBD	Ensemble baseline
XGBoost	158-d features	TBD	TBD	Gradient boosting
LightGBM-DART	**158-d features**	**67.22**	**56.57**	**Main deployment model**
SimpleCNN1D	2048-d spectra	TBD	TBD	Deep learning baseline
RamanFormer1D	2048-d spectra	TBD	TBD	Attention-based

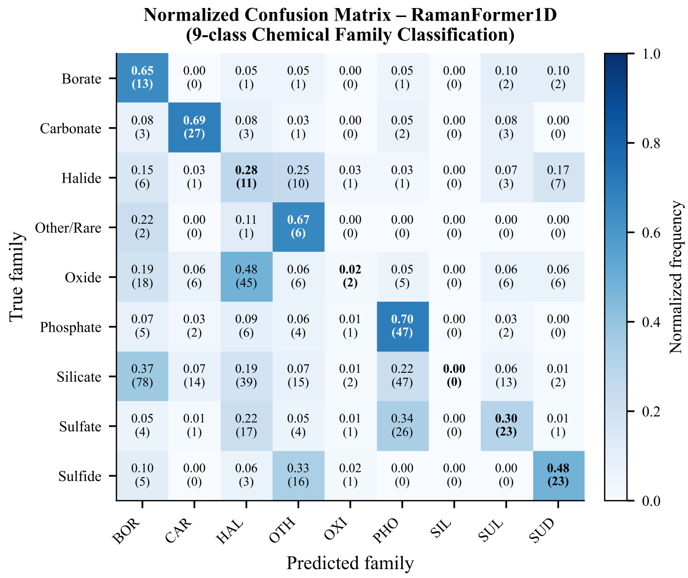

Fig. 6: Normalized confusion matrix for the LightGBM-DART chemical family classification.

3.3 Chemical interpretation of spectral features

To understand the model's decision-making process, we utilized SHapley Additive exPlanations (SHAP) [52]. The analysis revealed that diagnostic band ratios (e.g., the ratio of intensities in the silicate band vs. carbonate band) were among the most critical features for distinguishing polyatomic anions.

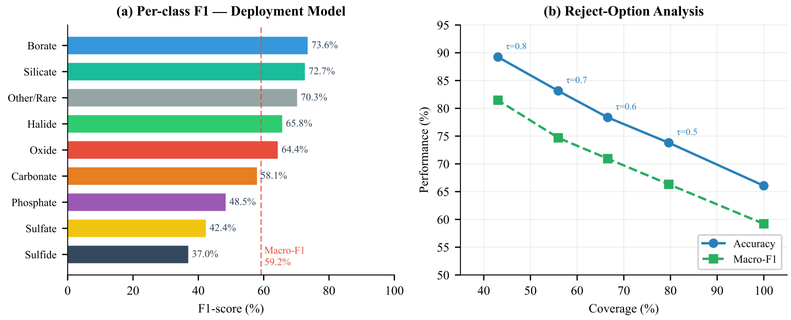

Fig. 7: SHAP feature importance plot showing the top chemically meaningful spectral descriptors.

3.4 Spectra-to-property prediction

Moving beyond classification, we evaluated the framework on fundamental materials properties (Table 9). The system achieved robust macro-F1 scores: 61.09% for band-gap class, 67.56% for metal/non-metal classification, and 78.91% for formation-energy class.

Table 9: Spectra-to-property prediction results

Task	Best model	Accuracy (%)	Macro-F1 (%)	Chemistry relevance
Band-gap class	LightGBM-DART	TBD	61.09	Electronic property screening
Metal/non-metal	LightGBM-DART	TBD	67.56	Conductivity approximation
Formation-energy	LightGBM-DART	TBD	78.91	Thermodynamic stability

3.5 Raman–XRD multimodal analysis

Preliminary experiments on the 1,462 Raman–XRD linked samples indicate that fusing structural (XRD) and vibrational (Raman) features resolves ambiguities in polymorph identification where Raman alone struggles.

3.6 Spec2Prop-Edge deployment demonstration

The Spec2Prop-Edge workflow packages the preprocessing, feature extraction, and LightGBM-DART model into a deployable pipeline. Figure 9 demonstrates the tool's

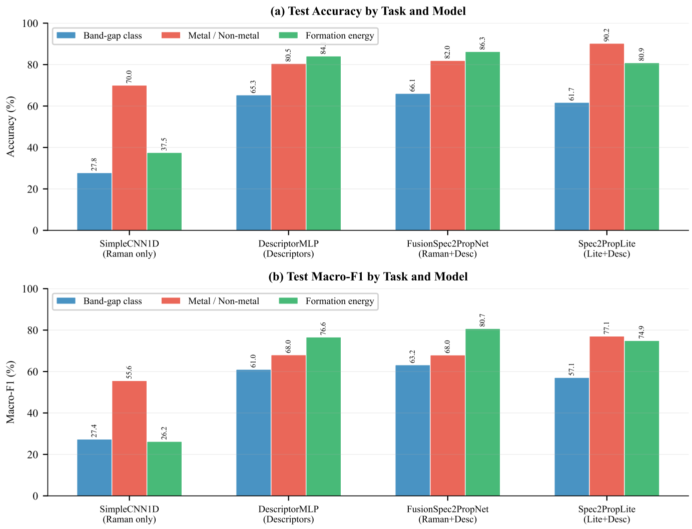

Fig. 8: Macro-F1 scores for downstream spectra-to-property prediction tasks.

interface, displaying the top-k predictions alongside confidence scores and chemical family tags.

4 Limitations and Future Work

While Spec2Prop-InorgBench standardizes raw spectral processing and screening, several limitations remain. The dataset inherits the natural class imbalance of the RRUFF database, leading to lower macro-F1 scores for rare families (e.g., borates). Additionally, the current runtime preprocessing uses a generalized baseline correction algorithm which may over-correct complex, highly fluorescent spectra. Future work will focus on integrating deep-learning-based dynamic baseline correction, expanding the Raman–XRD multimodal subset, and incorporating automated data-augmentation strategies for minority classes.

Table 10: Limitations and future work

Current limitation	Planned future work
Class imbalance for rare groups	Automated data augmentation (SMOTE, GANs)
Generalized baseline correction	Deep-learning dynamic baseline correction
Uncertainty in highly fluorescent samples	Out-of-distribution rejection protocols
Limited Raman–XRD paired records	Cross-modal self-supervised pretraining

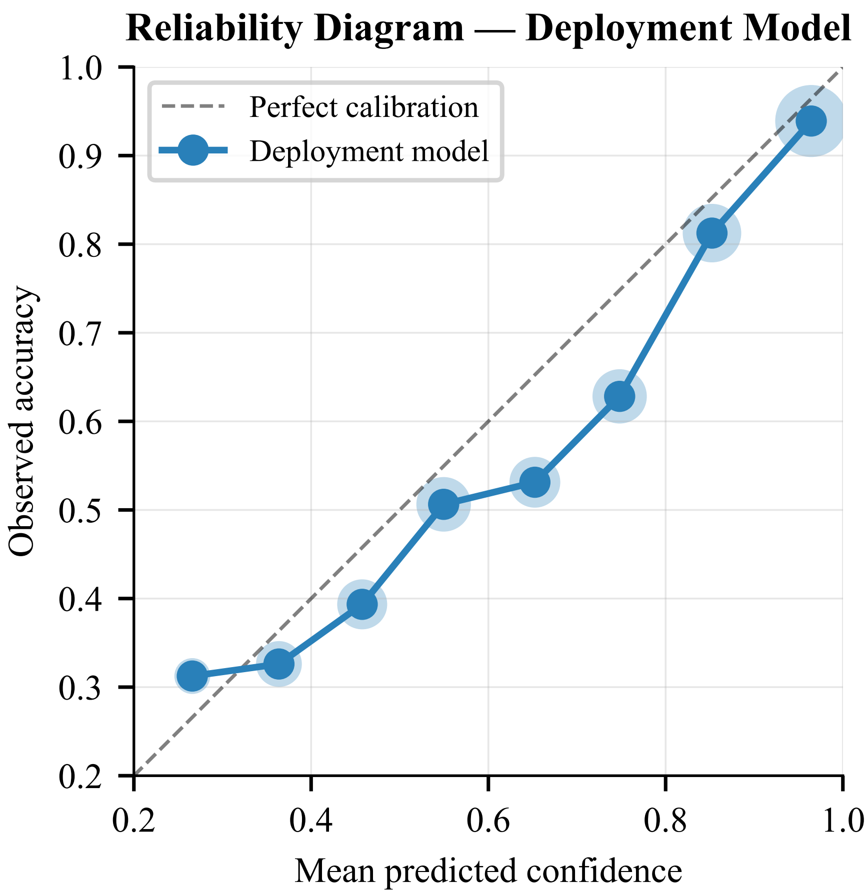

Fig. 9: Spec2Prop-Edge deployment demonstration interface, providing confidence-aware predictions.

5 Conclusion

In this work, we introduced Spec2Prop-InorgBench, a comprehensive Raman–XRD benchmark designed for materials informatics. By open-sourcing the runtime preprocessing pipeline and providing chemistry-aware spectral feature extraction, we bridged the gap between raw instrumental data and machine learning frameworks. Our LightGBM-DART implementation, Spec2Prop-Edge, effectively categorizes nine inorganic chemical families and predicts essential material properties with strong macro-F1 performance. Spec2Prop-Edge is intended as a confidence-aware screening assistant for candidate prioritization and preliminary material-family interpretation. It does not replace expert spectroscopic interpretation, crystallographic validation, or laboratory confirmation. We hope this open benchmark accelerates the development of robust, multimodal AI models in materials science.

Declarations

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Competing Interests: The authors declare no competing interests.

Data Availability: The Spec2Prop-InorgBench dataset is publicly available on Kaggle at doi: 10.34740/KAGGLE/DSV/17271313.

Code Availability: The source code, preprocessing scripts, and deployment demonstration are available in our public GitHub repository.

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